

Pengyue(Palmer) Zhao

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Summary

Software Engineer with experience building scalable backend systems, data pipelines, and AI automation workflows. Skilled in Java, Python, SQL, Spring Boot, and AWS, delivering reliable applications across fintech, e-commerce, and data-intensive environments.

Experience

Fincone

Los Angeles, United States

Software Engineer

04/2025 - Present

- Architected a RBAC authorization system using **Java Spring Boot**, securing 10,000+ users and reducing unauthorized access incidents by 97% through fine-grained permission modeling.
- Built a financial reporting service with **Redis** and optimized **SQL** queries, reducing reconciliation latency from 15 seconds to 2 seconds under concurrent transaction workloads.
- Integrated external payment gateways through **REST APIs** and XML protocols, adding validation, idempotency, and error handling to improve reliability and reduce transaction failures by 80% under high concurrency.

Pavilion Textile Group

Remote, Madrid, Spain

Technical Co-Founder

09/2024 - 02/2026

- Designed and implemented a full-stack ERP platform using **Django**, **React**, and **PostgreSQL**, reducing operational data inconsistencies by 95%.
- Applied **Cursor** and **Codex** to assist code generation, debugging, and documentation, fasten development by 50%.
- Automated inventory and product operations across 4 e-commerce platforms with Make.com, webhooks, and LLM workflows, eliminating 20+ hours of weekly manual reconciliation.
- Containerized and deployed services using **Docker** on **AWS EC2** and **RDS** with monitoring and environment isolation, achieving 99.9% system availability.

University of Amsterdam

Amsterdam, Netherlands

Data Engineer

09/2022 - 09/2023

- Engineered a NLP data pipeline using **Python** and **SQL** to process 1000+ institutional reports, extract ESG signals, and build **Tableau** analytics for research and data teams.

Education

University of Chicago

Chicago, United States

Master of Science in Computer Science

09/2023 - 03/2025

Rotterdam School of Management, Erasmus University

Rotterdam, Netherlands

Bachelor of Science in International Business Administration, 4.00 GPA

09/2020 - 07/2023

Technical Skills

Programming Languages: Python, Java, C++, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Django, React, Node.js, REST APIs

AI & Data: Cursor, Codex, LLM Workflows, NoSQL, Kafka, Spark, Hive, HBase, Redis, NLP, Tableau

Cloud & DevOps: AWS, Linux, Docker, Kubernetes, Git, CI/CD

Projects

Distributed Message Queue

Spring 2025

- Built a distributed message queue using **Python**, **REST APIs**, and Raft consensus, ensuring strong consistency and fault tolerance across 500 nodes with zero data loss.

Big Data Analytics Platform

Fall 2024

- Designed and implemented a Lambda Architecture system using **Kafka**, **Spark**, **Hive**, and **HBase**, enabling real-time data processing for a **Node.js** web application.

Genomics Annotation Cloud Service

Spring 2024

- Engineered a distributed, cloud-native platform on **AWS**, processing 1,000+ asynchronous jobs using event-driven architecture (**SQS/SNS**, **Lambda**) with auto-scaling and load balancing.